

Determine the region in the
z-plane

① $|z| \leq 3$

$$|z| \leq 3$$

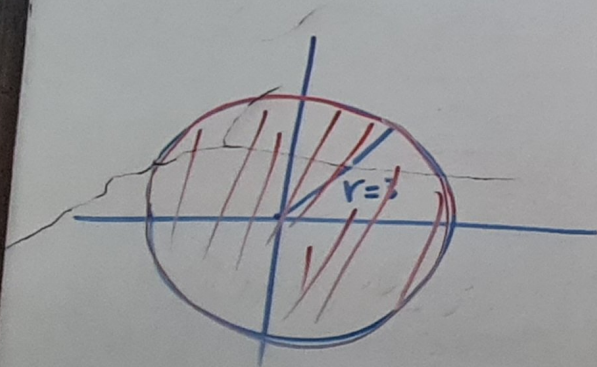
$$\sqrt{x^2 + y^2} \leq 3$$

$$x^2 + y^2 \leq 9$$

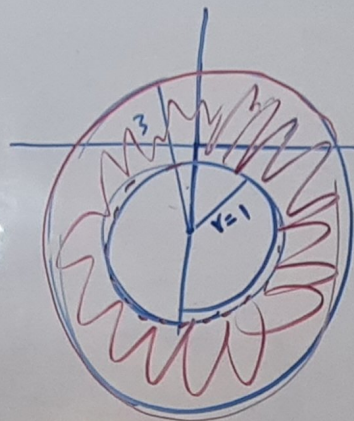
$$x^2 + y^2 = 9$$

$$r = 3$$

$$(0, 0)$$



② $1 < |z + 2i| \leq 3$



$$|z + 2i| > 1$$

$$r = 1$$

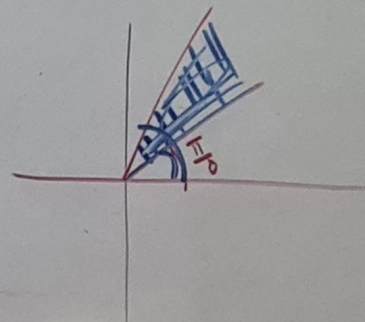
$$(0, -2)$$

$$|z + 2i| \leq 3$$

$$r = 3$$

$$(0, -2)$$

$$\frac{\pi}{6} \leq \arg(z) \leq \frac{\pi}{3}$$



$$w = f(z) = u + iv \quad (z = x + iy = r e^{i\theta})$$

write the function in the form $u + iv$

$$f(z) = z^2 - 2z + 1$$

$$\therefore z = x + iy$$

$$f(z) = (x + iy)^2 - 2(x + iy) + 1$$

$$= x^2 + 2ixy - y^2 - 2x - 2iy + 1$$

$$= (x^2 - y^2 - 2x + 1) + (2xy - 2y)i$$

u

v

$$② f(z) = |z|^2$$

$$|z|^2 = |x + iy|^2 = (\sqrt{x^2 + y^2})^2 = x^2 + y^2$$

$$③ f(z) = \frac{1}{z}$$

$$f(z) = \frac{1}{x + iy}$$

$$= \frac{1}{x + iy} \cdot \frac{x - iy}{x - iy}$$

$$= \frac{x - iy}{x^2 + y^2} = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}$$

$$f(z) = \ln(z) = \ln(r e^{i\theta})$$

$$= \ln(r) + \ln(e^{i\theta}) \quad (\ln e = 1)$$

$$= \ln(r) + i\theta$$

$$= \ln(\sqrt{x^2 + y^2}) + i \tan^{-1}\left(\frac{y}{x}\right) \quad \begin{matrix} r = \sqrt{x^2 + y^2} \\ \theta = \tan^{-1}\left(\frac{y}{x}\right) \end{matrix}$$

$$= \frac{1}{2} \ln(x^2 + y^2) + i \tan^{-1}\left(\frac{y}{x}\right)$$

u

v

$$w = f(z) = \underline{u} + i\underline{v} \quad (z = x + iy = r e^{i\theta})$$

write the function in the form
form of z

$$f(z) = \frac{x}{x^2 + y^2} - \frac{iy}{x^2 + y^2}$$

$$= \frac{x - iy}{x^2 + y^2}$$

$$= \frac{x - iy}{x^2 - y^2 i^2}$$

$$= \frac{x - iy}{(x - yi)(x + iy)}$$

$$= \frac{1}{x + iy}$$

$$= \frac{1}{z}$$

cauchy - riemann equations
کوتشې ریمان

$$f(z) = \underline{u(x, y)} + i \underline{v(x, y)}$$

$$(1) \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$(2) \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$\frac{\partial u}{\partial x}$ $\frac{\partial v}{\partial x}$
 $\frac{\partial u}{\partial y}$ $\frac{\partial v}{\partial y}$

u_x v_x
 u_y v_y

show that the cauchy riemann equations

... very point $f(z) = z^2 + z$

sol

$$f(z) = (x+iy)^2 + (x+iy)$$

$$= \underbrace{x^2 + 2ixy - y^2}_u + \underbrace{x+iy}_v$$

$$f(z) = \underbrace{(x^2 - y^2 + x)}_u + \underbrace{(2xy + y)}_v i$$

$$\frac{\partial u}{\partial x} = 2x + 1$$

$$\frac{\partial v}{\partial x} = 2y$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial y} = 2x + 1$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$w = f(z) = u + iv$$

$$\frac{dw}{dz} = f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

کفیت حساب مشتق باستخدام معادله
C-R eq

$$\frac{dz}{dz} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

or

$$= \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

$$\begin{matrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{matrix}$$

$f(z)$ is analytic functions
الالة تحليلية

$$\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$$

are continuous -- function
متواصلة
of x and y in R

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

If $w = \log(z)$

find $\frac{dw}{dz}$

and determine when w is non-analytic

Sol

$$w = \log(r \cdot e^{i\theta}) = \log(r) + \log e^{i\theta}$$

$$w = \log(r) + i\theta$$

$$w = \log(\sqrt{x^2+y^2}) + i \tan^{-1}\left(\frac{y}{x}\right)$$

$$w = \frac{1}{2} \log(x^2+y^2) + i \tan^{-1}\left(\frac{y}{x}\right)$$

$$u = \frac{1}{2} \log(x^2+y^2)$$

$$z = 0$$

$$= 0 + 0i$$

$$\frac{\partial u}{\partial x} = \frac{1}{2} \frac{2x}{x^2+y^2} = \frac{x}{x^2+y^2}$$

$$\frac{\partial v}{\partial x} = \frac{\left(\frac{y}{x}\right)'}{1 + \left(\frac{y}{x}\right)^2}$$

$$= \frac{-\frac{y}{x^2}}{1 + \frac{y^2}{x^2}} = \frac{-y}{x^2+y^2}$$

$$\frac{\partial u}{\partial y} = \frac{1}{2} \frac{2y}{x^2+y^2} = \frac{y}{x^2+y^2}$$

$$\frac{\partial v}{\partial y} = \frac{\left(\frac{y}{x}\right)'}{1 + \left(\frac{y}{x}\right)^2} = \frac{\frac{1}{x}}{1 + \frac{y^2}{x^2}}$$

$$\frac{\partial v}{\partial y} = \frac{x}{x^2+y^2}$$

$$\frac{dw}{dz} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$= \frac{x}{x^2+y^2} + i \frac{-y}{x^2+y^2}$$

$$= \frac{x-iy}{x^2+y^2} = \frac{x-iy}{(x-iy)(x+iy)} = \frac{1}{x+iy} = \frac{1}{z}$$

$$\frac{\partial}{\partial y} \left(\frac{y}{x} \right)$$

$$= \frac{1}{x} \frac{\partial}{\partial y} (y) = \frac{1}{x} (1) = \frac{1}{x}$$

$$\frac{\partial}{\partial x} (y^2)$$

$$= y^2 \frac{\partial}{\partial x} (1)$$

$$= y(0) = 0$$

$$\frac{\partial}{\partial y} (x^2)$$

$$= x^2 \frac{\partial}{\partial y} (1)$$

$$= x^2(0) = 0$$

$$\left(\tan^{-1}\left(\frac{y}{x}\right) \right)' = \frac{\frac{1}{x}}{1 + \left(\frac{y}{x}\right)^2}$$

$$\frac{\partial}{\partial x} \left(\frac{y}{x} \right) = y \frac{\partial}{\partial x} \left(\frac{1}{x} \right)$$

$$y \left(\frac{-1}{x^2} \right) = -\frac{y}{x^2}$$